Quadratics & Polynomials

Section 1: Factoring & Quadratic Formula

1 What is the positive solution to the quadratic equation $3x^2 - 10x - 8 = 0$?

a. 2

b. 4

c. $\frac{2}{3}$

d. $\frac{4}{3}$

2 If $x^2 - 6x + k = 0$ has two equal real roots, what is the value of k?

a. 3

b. 6

c. 9

d. 36

3 What is the sum of the solutions of the quadratic equation $2x^2 + 8x - 15 = 0$?

a. -4

b. 4

c. -7.5

d. 8

Section 2: Vertex Form & Parabola Extremes

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4 What is the minimum value of the quadratic function $f(x) = 2x^2 - 12x + 23$?

a. 3

b. 5

c. 7

d. 11

5 The parabola $y = -(x - 4)^2 + 16$ models the flight path of a projectile. What is the maximum height achieved?

a. 4

b. 8

c. 16

d. 32

6 What is the axis of symmetry of the parabola $y = x^2 + 8x - 5$?

a. $x = -4$

b. $x = 4$

c. $x = -8$

d. $y = -4$

Section 3: Polynomial Division & Factors

7 What is the remainder when $P(x) = x^3 - 4x^2 + 5x - 7$ is divided by $(x - 2)$?

a. -5

b. -3

c. 3

d. 5

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8 If $(x-3)$ is a factor of $x^3 - 2x^2 - kx + 12$, what is the value of k?

a. 3

b. 4

c. 7

d. 9

9 Which of the following polynomials has roots at $x = -1, x = 2,$ and $x = 3$?

a. $x^3 - 4x^2 + x + 6$

b. $x^3 - 4x^2 - x + 6$

c. $x^3 + 4x^2 + x - 6$

d. $x^3 - 2x^2 - 5x + 6$

Section 4: Advanced Polynomial Graphs & Vieta

10 What are all real roots of the polynomial equation $x^4 - 5x^2 + 4 = 0$?

a. $x = \pm 1, \pm 2$

b. $x = \pm 1, \pm 4$

c. $x = 1, 2$

d. $x = 0, 1, 4$

11 What is the product of the roots of the cubic equation $2x^3 - 6x^2 + 4x - 10 = 0$?

a. 2

b. 3

c. 5

d. -5

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12 How many times does the graph of $f(x) = (x - 1)^2(x + 3)$ cross the x-axis?

a. 1 time (crosses at $x = -3$, touches at $x = 1$)

b. 2 times

c. 3 times

d. 0 times

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